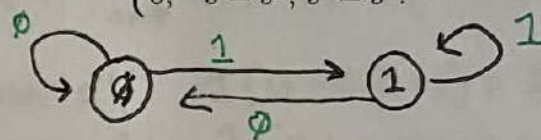


Problem 1

Consider a system with two states $s_k \in \{s^1 = 0, s^2 = 1\}$. There are two possible actions: a^1 and a^2 . The transition probabilities can be written as:

$$p(s' | s, a^1) = \begin{cases} 1, & s = s^1, s' = s^1, \\ 0, & s = s^1, s' = s^2, \\ 0, & s = s^2, s' = s^1, \\ 1, & s = s^2, s' = s^2, \end{cases} \quad p(s' | s, a^2) = \begin{cases} 0, & s = s^1, s' = s^1, \\ 1, & s = s^1, s' = s^2, \\ 1, & s = s^2, s' = s^1, \\ 0, & s = s^2, s' = s^2. \end{cases}$$

Reward Function: $R(s, a, s') = \begin{cases} 1, & s' = s^2 \\ 0, & s' = s^1 \end{cases}$



Start with a random policy: $\pi^0(s^1) = a^1$, $\pi^0(s^2) = a^2$, $\gamma = 0.8$, $\theta = 0.9$. Use Policy Iteration to compute $\pi^1(s^1)$, $\pi^1(s^2)$, with $V_0(s^1) = V_0(s^2) = 0$ for initialization of policy evaluation.

Recall Bellman's Equation: $v^{\pi}(s) = \sum p(s' | s, \pi(s)) [R(s, \pi(s), s') + \gamma v^{\pi}(s')]$

Policy Evaluation: π^0

$$0 = v^{\pi^0}(s^1) = 1 [0 + 0.8 (v^{\pi^0}(s^1))]$$

$$0 = .8 (v^{\pi^0}(s^1)) \rightarrow v^{\pi^0}(s^1) = 0$$

$$V^{\pi^0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$0 = v^{\pi^0}(s^2) = 1 (0 + 0.8 v^{\pi^0}(s^2)) \rightarrow v^{\pi^0}(s^2) = 0$$

Policy Improvement: $\max_{a \in A} Q(s, a) = \sum p(s' | s, a) [R(s, a, s') + \gamma V^{\pi}(s')]$

At $s^1 \rightarrow Q(s^1, a^1) = 1 [0 + 0.8(0)] = 0$

$$Q(s^1, a^2) = 1 [1 + 0.8(0)] = 1$$

$$\arg \max_{a \in A} (Q(s^1, a^1), Q(s^1, a^2)) = Q(s^1, a^2)$$

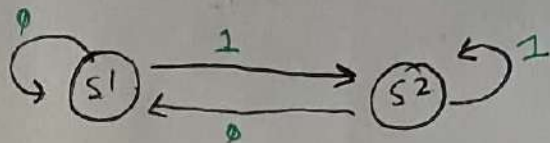
$$\boxed{\pi^1(s^1) = a^2}$$

At $s^2 \rightarrow Q(s^2, a^1) = 1 [1 + 0.8(0)] = 1$

$$Q(s^2, a^2) = 1 [0 + 0.8(0)] = 0$$

$$\arg \max_{a \in A} (Q(s^2, a^1), Q(s^2, a^2)) = Q(s^2, a^1)$$

$$\boxed{\pi^1(s^2) = a^1}$$



Problem 2

Consider the problem defined in Problem 1.

- (a) Given $\begin{pmatrix} V_0(s^1) \\ V_0(s^2) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and $\gamma = 0.8$, perform the **Value Iteration** method to compute V_1, V_2, V_3 .

- (b) Compute $\pi(s = s^1)$ and $\pi(s = s^2)$ associated with V_3 .

a) Recall Bellman Optimality: $V_{k+1}(s) = \max_{a \in A} \sum_{s'} p(s'|s, a) [R(s, a, s') + \gamma V_k(s')]$

Iteration 1: For state s^1 , taking action a^1 : $1[0 + 0.8(0)] = 0$
 taking action a^2 : $1[1 + 0.8(0)] = 1$

$$V_1(s^1) = 1$$

For state s^2 , taking action a^1 : $1[1 + 0.8(0)] = 1$
 taking action a^2 : $1[0 + 0.8(0)] = 0$ $V_1(s^2) = 1$

$$V_1 = (1, 1)$$

Iteration 2: s^1, a^1 : $1[0 + 0.8(1)] = 0.8$ s^1, a^2 : $1[1 + 0.8(1)] = 1.8$
 ~~s^1, a^1~~ s^2, a^1 : $1[1 + 0.8(1)] = 1.8$ s^2, a^2 : $1[0 + 0.8(1)] = 0.8$

$$V_2 = (1.8, 1.8)$$

Iteration 3: s^1, a^1 : $1[0 + 0.8(1.8)] = 1.44$ $\Rightarrow 2.44 = V_3(s^1)$
 s^1, a^2 : $1[1 + 0.8(1.8)] = 2.44$
 s^2, a^1 : $1[1 + 0.8(1.8)] = 2.44$ $\Rightarrow 2.44 = V_3(s^2)$
 s^2, a^2 : $1[0 + 0.8(1.8)] = 1.44$

$$V_3 = (2.44, 2.44)$$

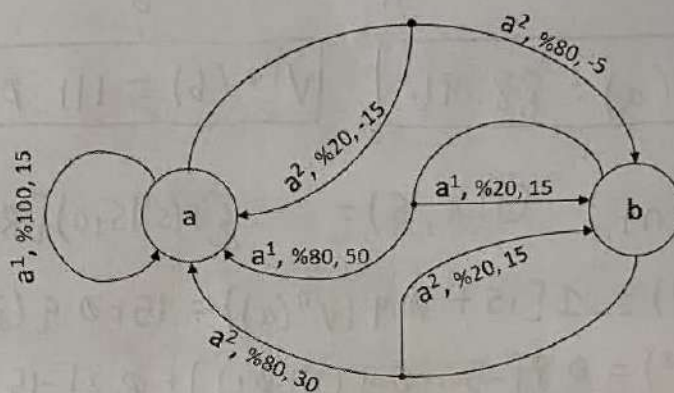
b) At V_3 , $a^1 = 1.44$ and $a^2 = 2.44$ for $s^1 \rightarrow \max(a^1, a^2) = a^2$
 $\pi(s = s^1) = a^2 \rightarrow \text{Switch}$

At V_3 , for s^2 , $a^1 = 2.44$ and $a^2 = 1.44 \rightarrow \max(a^1, a^2) = a^1$
 $\pi(s = s^2) = a^1 \rightarrow \text{stay}$

Problem 3

Consider the following MDP having two states: a, and b. In each state, there are two possible actions: a^1 , and a^2 . The transition model and reward are shown in the diagram below. Apply Policy Iteration to determine the optimal policy and state values of a and b. Assume the initial policy is action a^2 for both states, $\gamma = 0.9$. For evaluation of policy, you need to solve two set of linear equations for the following form, instead of iterative steps of policy evaluation:

$$V^\pi(s) = \sum_{s', r} P(s' | s, \pi(s)) [R(s, \pi(s), s') + \gamma V^\pi(s')].$$



Here is an example of transition and reward from the diagram: In state a, action a^2 moves the agent to state b with probability 0.8 with the corresponding reward -5, and make the agent stay at state a with probability 0.2 and corresponding reward -15.

Recall Bellman's Equation: $V^\pi(s) = \mathbb{E} P(s' | s, \pi(s)) [R(s, \pi(s), s') + \gamma V^\pi(s')]$

Policy Evaluation: $V^\pi(a) = 0.8 [-5 + 0.9(V^\pi(b))] + 0.2 [-15 + 0.9(V^\pi(a))]$

given initial policy a^2 $V^\pi(b) = 0.8 [30 + 0.9V^\pi(a)] + 0.2 [15 + 0.9(V^\pi(b))]$

$$V^\pi(a) = -4 + 0.72 V^\pi(b) - 3 + 0.18 V^\pi(a)$$

$$V^\pi(b) = 24 + 0.72 V^\pi(a) + 3 + 0.18 V^\pi(b)$$

$$V^\pi(a) = -7 + 0.18 V^\pi(a) - 0.72 V^\pi(b)$$

$$V^\pi(b) = 27 + 0.72 V^\pi(a) + 0.18 V^\pi(b)$$

2 equations
2 unknowns

$$\begin{cases} 7 = -0.82 V^\pi(a) + 0.72 V^\pi(b) \\ -27 = 0.72 V^\pi(a) + 0.18 V^\pi(b) \end{cases}$$

$$\begin{aligned} \textcircled{1} \quad & \begin{bmatrix} -0.82 & 0.72 \\ 0.72 & -0.82 \end{bmatrix} \begin{bmatrix} v^\pi(a) \\ v^\pi(b) \end{bmatrix} = \begin{bmatrix} 7 \\ -27 \end{bmatrix} \\ \textcircled{2} \quad & \begin{matrix} A & X & b \end{matrix} \end{aligned}$$

$$A^{-1}b = x, \text{ so } A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{0.6724 - 0.5184} \begin{bmatrix} -0.82 & -0.72 \\ -0.72 & -0.82 \end{bmatrix}$$

$$A^{-1}b = \begin{bmatrix} -5.32 & -4.675 \\ -4.675 & -5.32 \end{bmatrix} \begin{bmatrix} 7 \\ -27 \end{bmatrix} = \begin{bmatrix} 88.96 \\ 111.04 \end{bmatrix}$$

$A^{-1} \quad b \quad x$

$$V^{\pi^0}(a) = 88.96$$

$$V^{\pi^0}(b) = 111.04$$

Policy Improvement: $Q(a, s) = \sum p(s'|s, a) [R(s, a, s') + \gamma V^{\pi}(s')]$

State a , action a^1 : $Q(a, a^1) = 1[15 + 0.9(V^{\pi^0}(a))] = 15 + 0.9(88.96) = 95.064$

$Q(a, a^2) = 0.8[-5 + 0.9(111.04)] + 0.2[-15 + 0.9(88.96)] = 88.96$

$\max_{a \in A} [Q(a, a^1), Q(a, a^2)] = Q(a, a^1), \text{ choose } a^1$

$$\pi^1(a) = a^1$$

~~$$\pi^1(a) = a^2$$~~

$Q(b, a^1) = 0.8[50 + 0.9(88.96)] + 0.2[15 + 0.9(111.04)] = 127.04$

$Q(b, a^2) = 0.8[30 + 0.9(88.96)] + 0.2[15 + 0.9(111.04)] = 111.04$

$\max_{a \in A} [Q(b, a^1), Q(b, a^2)] = Q(b, a^1)$

$$\pi^1(b) = a^1$$

~~$$\pi^1(b) = a^2$$~~

choose action a^1

Since $\pi^1 \neq \pi^0$, need to iterate again

Iteration 2 $\rightarrow$ Policy Evaluation $\pi_1 \xrightarrow{PE} v^{\pi_1}$

under $\pi_1 = \begin{bmatrix} a^1 \\ a^1 \end{bmatrix}$

$$V^{\pi}(s) = \sum p(s'|s, \pi(s)) [R(s, \pi(s), s') + \gamma V^{\pi}(s')]$$

$$\begin{cases} V^{\pi_1}(a) = 1[15 + 0.9 V^{\pi_1}(a)] \rightarrow \boxed{V^{\pi_1}(a) = 150} \\ V^{\pi_1}(b) = 0.2[15 + 0.9 V^{\pi_1}(b)] + 0.8[50 + 0.9 V^{\pi_1}(a)] \end{cases}$$

$$V^{\pi_1}(b) = 3 + 0.18 V^{\pi_1}(b) + 148 \rightarrow \boxed{V^{\pi_1}(b) = 184.15}$$

$$V^{\pi_1} = \begin{bmatrix} 150 \\ 184.15 \end{bmatrix}$$

Policy Improvement: $v^{\pi_1} \xrightarrow{PI} \pi_2$

$$V^*(s) = \max_{a \in A} Q^*(s, a)$$

$$Q(a, a^1) = 1[15 + 0.9(150)] = 150 \quad \nearrow \max(Q(a, a^1), Q(a, a^2)) = a^2$$

$$Q(a, a^2) = 0.8[-5 + 0.9(184.15)] + 0.2[-15 + 0.9(150)] = 128.6 + 24 = \boxed{152.59}$$

$$Q(b, a^1) = 0.8[50 + 0.9(150)] + 0.2[15 + 0.9(184.15)] = 148 + 36.15$$

$$Q(b, a^2) = 0.8[30 + 0.9(150)] + 0.2[15 + 0.9(184.15)] = \boxed{184.15}$$

$$= 132 + 36.15 = 168.15$$

$$\boxed{\begin{matrix} \pi^2(a) = a^2 \\ \pi^2(b) = a^1 \end{matrix}}$$

$$Q(b, a^1) > Q(b, a^2) \quad \text{choose } a^1$$

$$\pi^2 = \begin{bmatrix} a^2 \\ a^1 \end{bmatrix}$$

But $\pi^2 \neq \pi^1$ so need to iterate again

Iteration 3 $\rightarrow$ Policy Evaluation $\pi_2 \rightarrow v^{\pi_2}$

$$V^{\pi}(s) = \sum p(s'|s, \pi(s)) [R(s, \pi(s), s') + \gamma V^{\pi}(s')]$$

under $\pi_2 = \begin{bmatrix} a^2 \\ a^1 \end{bmatrix}$

$$V^{\pi_2}(a) = 0.8[-5 + 0.9(V^{\pi_2}(b))] + 0.2[-15 + 0.9(V^{\pi_2}(a))]$$

$$V^{\pi_2}(b) = 0.8[50 + 0.9(V^{\pi_2}(a))] + 0.2[15 + 0.9 V^{\pi_2}(b)]$$

$$7 = -0.82 V(a) + 0.72 V(b)$$

$$-43 = 0.72 V(a) - 0.82 V(b)$$

$$\begin{matrix} \begin{bmatrix} -0.82 & 0.72 \\ 0.72 & -0.82 \end{bmatrix} & \begin{bmatrix} V(a) \\ V(b) \end{bmatrix} & = & \begin{bmatrix} 7 \\ -43 \end{bmatrix} \\ A & X & & b \end{matrix}$$

$$A^{-1} = \frac{1}{0.154} \begin{bmatrix} -0.82 & -0.72 \\ -0.72 & -0.82 \end{bmatrix} = \begin{bmatrix} -5.325 & -4.675 \\ -4.675 & -5.325 \end{bmatrix}$$

$$A^{-1}b = X \rightarrow \begin{bmatrix} -5.325 & -4.675 \\ -4.675 & -5.325 \end{bmatrix} \begin{bmatrix} 7 \\ -43 \end{bmatrix} = \begin{bmatrix} 163.77 \\ 196.23 \end{bmatrix}$$

$$V^{\pi_2} = \begin{bmatrix} 163.77 \\ 196.23 \end{bmatrix}$$

Policy Improvement: $V^{\pi_2} \xrightarrow{PI} \pi_3$ $V^*(s) = \max_{a \in A} Q^*(s, a)$

$$Q(a, a^1) = 1[15 + 0.9(163.77)] = 162.39$$

$$Q(a, a^2) = 0.8[-5 + 0.9(196.23)] + 0.2[-15 + 0.9(163.77)] = 137.29 + 26.48$$

$$Q(a, a^2) > Q(a, a^1) \rightarrow \pi^3(a) = a^2 = 163.77$$

$$Q(b, a^1) = 0.8[15 + 0.9(163.77)] + 0.2[15 + 0.9(196.23)] = 157.91 + 38.32 = 196.23$$

$$Q(b, a^2) = 0.8[30 + 0.9(163.77)] + 0.2[15 + 0.9(196.23)] = 141.91 + 38.32 = 180.23$$

$$Q(b, a^1) > Q(b, a^2) \rightarrow \pi^3(b) = a^1$$

$\pi^3 = \pi^2$, No change so is optimal policy

$$\begin{matrix} \pi^*(a) = a^2 \\ \pi^*(b) = a^1 \end{matrix}$$

$$\begin{matrix} V^*(a) = 163.77 \\ V^*(b) = 196.23 \end{matrix}$$

Problem 4

Consider the following maze with 14 states and a goal. The agent can take one of the following four actions at any given state $A = \{Up, Down, Right, Left\}$. The state transitions are deterministic; for example, $P(s' = 10 \mid s = 12, a = Up) = 1$. The reward is as follows:

$$\text{Reward}(s, a, s') = \begin{cases} -1, & \text{if taking any action,} \\ +100, & \text{if moving to goal,} \\ -5, & \text{if moving to bump.} \end{cases}$$

- Using $\gamma = 1$ and $\theta = 0.6$, perform vector-form **Value Iteration** method with $V_0(s) = 0$ to compute V^* .
- Compute the optimal policy.

**** Show all intermediate state values in maze, without details of calculation. ****

	7	8	9	10
6				11
5				
4				G
	3	S	2	1

See Next page

V_0

$\emptyset_7$	$\emptyset_8$	$\emptyset_9$	$\emptyset_{10}$
$\emptyset_6$		$\emptyset_{11}$	$\emptyset_{12}$
$\emptyset_5$		$\emptyset_{13}$	$\emptyset_{14}$
$\emptyset_4$			
$\emptyset_3$	$\emptyset_2$	$\emptyset_1$	

V_1

-1_7	-1_8	-1_9	-1_{10}
-1_6		-1_{11}	-1_{12}
-1_5		-1_{13}	99_{14}
-1_4			
-1_3	-1_2	-1_1	

V_2

-2_7	-2_8	-2_9	-2_{10}
-2_6		-2_{11}	93_{12}
-2_5		93_{13}	99_{14}
-2_4			
-2_3	-2_2	-2_1	

V_3

-3_7	-3_8	-3_9	92_{10}
-3_6		92_{11}	93_{12}
-3_5		93_{13}	99_{14}
-3_4			
-3_3	-3_2	-3_1	

V_4

-4_7	-4_8	91_9	92_{10}
-4_6		92_{11}	93_{12}
-4_5		93_{13}	99_{14}
-4_4			
-4_3	-4_2	-4_1	

V_5

-5_7	90_8	91_9	92_{10}
-5_6		92_{11}	93_{12}
-5_5		93_{13}	99_{14}
-5_4			
-5_3	-5_2	-5_1	

V₆

89 ₇	90 ₈	91 ₉	92 ₁₀
-6 ₆		92 ₁₁	93 ₁₂
-6 ₅		93 ₁₃	99 ₁₄
-6 ₄			
-6 ₃	-6 ₂	-6 ₁	

V₇

89 ₇	90 ₈	91 ₉	92 ₁₀
88 ₆		92 ₁₁	93 ₁₂
-7 ₅		93 ₁₃	99 ₁₄
-7 ₄			
-7 ₃	-7 ₂	-7 ₁	

V₈

89 ₇	90 ₈	91 ₉	92 ₁₀
88 ₆		92 ₁₁	93 ₁₂
87 ₅		93 ₁₃	99 ₁₄
-8 ₄			
-8 ₃	-8 ₂	-8 ₁	

V₉

89 ₇	90 ₈	91 ₉	92 ₁₀
88 ₆		92 ₁₁	93 ₁₂
87 ₅		93 ₁₃	99 ₁₄
86 ₄			
-9 ₃	-9 ₂	-9 ₁	

V₁₀

89 ₇	90 ₈	91 ₉	92 ₁₀
88 ₆		92 ₁₁	93 ₁₂
87 ₅		93 ₁₃	99 ₁₄
86 ₄			
85 ₃	-10 ₂	-10 ₁	

V₁₁

89 ₇	90 ₈	91 ₉	92 ₁₀
88 ₆		92 ₁₁	93 ₁₂
87 ₅		93 ₁₃	99 ₁₄
86 ₄			
85 ₃	79 ₂	-11 ₁	

V_{12}

89 ₇	90 ₈	91 ₉	92 ₁₀
88 ₆		92 ₁₁	93 ₁₂
87 ₅		93 ₁₃	99 ₁₄
86 ₄			
85 ₃	79 ₂	78 ₁	

V_{13}

89 ₇	90 ₈	91 ₉	92 ₁₀
88 ₆		92 ₁₁	93 ₁₂
87 ₅		93 ₁₃	99 ₁₄
86 ₄			
85 ₃	79 ₂	78 ₁	

$$\|V_{13} - V_{12}\| < \theta$$

$$\text{SO } V_{13} = V^*$$

To find π^*

~~maximize~~ Up D R L

$$\pi^*(14) = [92, 99, 93, 92]$$

$$\pi^*(13) = [86, 92, 93, 92]$$

$$\pi^*(12) = [91, 93, 92, 86]$$

$$\pi^*(11) = [90, 92, 92, 91]$$

$$\pi^*(10) = [91, 92, 91, 90]$$

$$\pi^*(9) = [90, 86, 91, 89]$$

$$\pi^*(8) = [89, 89, 90, 88]$$

$$\pi^*(7) = [88, 87, 89, 88]$$

$$\pi^*(6) = [88, 86, 87, 87]$$

$$\pi^*(5) = [87, 85, 86, 86]$$

$$\pi^*(4) = [86, 84, 85, 85]$$

$$\pi^*(3) = [85, 79, 78, 79]$$

$$\pi^*(2) = [78, 78, 77, 79]$$

$$\pi^*(1) = [77, 77, 77, 78]$$

→ ₇	→ ₈	→ ₉	↓ ₁₀
↑ ₆		↘ ₁₁	↓ ₁₂
↑ ₅		→ ₁₃	↓ ₁₄
↑ ₄			
↑ ₃	← ₂	← ₁	

Problem 5

For an MDP defined by state space $\mathcal{S}$, action space $\mathcal{A}$, reward $R(s, a, s')$, and transition probability $p(s' | s, a)$, write the following:

(a) For a given policy π , write

- $V^\pi(s)$ based on $V^\pi \rightarrow V^\pi(s) = \sum p(s' | s, \pi(s)) [R(s, \pi(s), s') + \gamma V^\pi(s')]$
- $V^\pi(s)$ based on $Q^\pi \rightarrow V^\pi(s) = \sum_a Q^\pi(s, \pi(s))$
- $Q^\pi(s, a)$ based on $V^\pi \rightarrow Q^\pi(s, a) = \sum p(s' | s, a) [R(s, a, s') + \gamma V^\pi(s')]$
- $Q^\pi(s, a)$ based on $Q^\pi \rightarrow Q^\pi(s, a) = \sum p(s' | s, a) [R(s, a, s') + \gamma Q^\pi(s', a')]$

(b) For the optimal policy π^* , write

- $V^*(s)$ based on $V^* \rightarrow V^*(s) = \max_{a \in \mathcal{A}} \sum p(s' | s, a) [R(s, a, s') + \gamma V^*(s')]$
- $V^*(s)$ based on $Q^* \rightarrow V^*(s) = \max_{a \in \mathcal{A}} Q^*(s, a)$
- $Q^*(s, a)$ based on $V^* \rightarrow Q^*(s, a) = \sum p(s' | s, a) [R(s, a, s') + \gamma V^*(s')]$
- $Q^*(s, a)$ based on $Q^* \rightarrow Q^*(s, a) = \sum p(s' | s, a) [R(s, a, s') + \gamma \max_{a' \in \mathcal{A}} Q^*(s', a')]$

Help: An example of response:

$$V^\pi(s) = \sum_{s'} p(s' | s, \pi(s)) [R(s, \pi(s), s') + \gamma V^\pi(s')].$$

Questions about the HW should be directed to TA, Amirhossein Ravari, at: ravari.a@northeastern.edu